FIELD JOURNAL

# Hidden Markov Models

*Infer hidden states with visible actions to inferred paths and uncertainty*

## Chapter I | Understanding the Hidden Process

*Model the unobserved customer journey and how it generates the actions we see.*

### 1. What is a hidden state?

- At each step, a customer has an unobserved decision state (e.g., Browsing or Evaluating) that influences the visible action (e.g., Search, View, Compare).
- The same action can arise from different hidden states.

*Example: A View action might reflect casual Browsing (e.g., looking around) or deliberate Evaluating (e.g., comparing options). The hidden explanation is uncertain.*

### 2. Markov chain vs. HMM

*Why not model the actions directly?*

**Markov chain (actions only)**
*Models how one observed action leads to the next.*

Search → View

**Hidden Markov Model (HMM)**
*Adds hidden states that evolve over time and generate the actions we observe.*

B → E

Search, View

### 3. The HMM: a two-layer process

*At each step $t_i$ a hidden state generates an action.*

Hidden state (intent): B → E → E → E

Observed action, (evidence): Search, View, Compare, View

$t_1$ $t_2$ $t_3$ $t_4$

- **Horizontal arrows:** state transitions $P(H_i \mid H_{i-1})$
- **Vertical arrows:** emissions $P(O_i \mid H_i)$

*Note: This shows one possible hidden path (B → E → E → E) that could generate the observed actions.*

### 4. Givens and unknowns

*What we assume is known, and what we infer.*

| | |
|---|---|
| Known (model inputs) | $\pi$ Initial state probabilities: $P(H_1 = s)$<br>A Transition probabilities: $P(H_i = s' \mid H_{i-1} = s)$<br>B Emission probabilities: $P(O_i = o \mid H_i = s)$ |
| O | Observed actions/evidence: $O = S \to V \to C \to V$ |
| **Unknown (to infer)** | • The hidden state sequence $H_{1:T}$ (e.g., $B \to E \to E \to E$).<br>• Probabilities about states and paths (e.g., $P(H_4 = s \mid O)$, $P(H_{1:T} \mid O)$). |

### 5. The Generative Model: How an HMM Creates a Journey

*The model is defined by three probability distributions.*

| | | |
|---|---|---|
| $\pi$ (initial) | $P(H_1 = s)$ | Where the journey begins (starting state). |
| A (transition) | $P(H_i = s' \mid H_{i-1} = s)$ | How the hidden state changes between steps. |
| B (emission) | $P(O_i = o \mid H_i = s)$ | Which action a state produces (at each step). |

**Initial $\pi$**

| s | $P(H_1 = s)$ |
|---|---|
| B | 0.80 |
| E | 0.20 |

**Transition A**

| s \ s' | B | E |
|---|---|---|
| B | 0.80 | 0.20 |
| E | 0.20 | 0.80 |

**Emission B**

| s \ o | Search | View | Compare |
|---|---|---|---|
| B | 0.50 | 0.30 | 0.20 |
| E | 0.10 | 0.40 | 0.50 |

**How to read:**

- $\pi(B) = 0.80$: start in *Browsing* with probability 0.80.
- $A(B, E) = 0.20$: *move from Browsing to Evaluating* with probability 0.20.
- $B(E, View) = 0.40$: *if in Evaluating, emit View* with probability 0.40.

### 6. Generating one journey: a worked example

*Follow one possible hidden path and see how its probability is computed.*

Hidden path $H_{1:T}$: B → E → E → E

Observed actions $O_{1:T}$: Search, View, Compare, View

$t_1$ $t_2$ $t_3$ $t_4$

**Joint probability of this path and these actions**

$t_1$: start + emit: $\pi(B) \times B(B, Search) = 0.80 \times 0.50 = 0.40$

$t_2$: move + emit: $A(B, E) \times B(E, View) = 0.20 \times 0.40 = 0.08$

$t_3$: stay + emit: $A(E, E) \times B(E, Compare) = 0.80 \times 0.50 = 0.40$

$t_4$: stay + emit: $A(E, E) \times B(E, View) = 0.80 \times 0.40 = 0.32$

$$P(H_{1:T}, O_{1:T}) = \pi(B) \times B(B, Search) \times A(B, E) \times B(E, View) \times A(E, E) \times B(E, Compare) \times A(E, E) \times B(E, View)$$
$$= 0.80 \times 0.50 \times 0.20 \times 0.40 \times 0.80 \times 0.50 \times 0.80 \times 0.40$$
$$= \mathbf{0.004096} \quad (= 0.4096\%)$$

*Read the worked path from left to right: first start and emit the initial action, then at each later step transition to the next hidden state and emit that step's observation.*

**Step-by-step generation** *(with numbers)*

1. **Start:** $H_1 = B$, $\pi(B) = 0.80$; emit *Search*, $B(B, Search) = 0.50$.
2. **Transition:** $B \to E$, $A(B, E) = 0.20$; emit *View*, $B(E, View) = 0.40$.
3. **Transition:** $E \to E$, $A(E, E) = 0.80$; emit *Compare*, $B(E, Compare) = 0.50$.
4. **Transition:** $E \to E$, $A(E, E) = 0.80$; emit *View*, $B(E, View) = 0.40$.

**Why we can multiply**
*(the assumptions)*

- *Markov property:* the next hidden state depends only on the previous hidden state, not the entire past.
- *Emission independence:* once the current hidden state is fixed, the emitted action depends only on that current state (not on earlier actions or states).

**What this means**

- $0.004096$ is the joint probability of this specific hidden path ($B \to E \to E \to E$) together with the observed actions (Search, View, Compare, View).
- It is not the total probability of the observations, because other hidden paths could also generate Search, View, Compare, View.
- To get that total, sum the joint probabilities of all possible hidden paths to get the total probability of the observations.

## Chapter II — *Performing Inference with Dynamic Programming*

*Use efficient algorithms to answer key questions about the hidden journey.*

**Why dynamic programming?**

Many hidden paths can produce the same observed sequence, so brute-force enumeration grows quickly (exponentially) with sequence length.

**Big idea:** Reuse overlapping subproblems instead of recomputing every path.

*For this example: With 2 hidden states over 4 time steps, there are $2^4 = 16$ possible hidden paths.*

**Takeaway:** *Turn an exponential search into an efficient, dynamic program.*

**The trellis: shared routes, reused results.**

B and E—two hidden states. Reusing results from $t_2$ to $t_3$.

B = Browsing
E = Evaluating
$t$ = position (time step)

B: 0.400 → 0.0972 → 0.0171 → 0.006
E: 0.020 → 0.0384 → 0.0225 → 0.009
$t_1$ $t_2$ $t_3$ $t_4$

**Key idea:** At each state-position, we keep one summary of everything that matters so far, and reuse it for all paths that arrive there.

**Takeaway:** *Forward stores a sum (total probability of all arriving paths); Viterbi stores a max (the single most likely arriving path).*

**Three inference questions.**

*Different computations on the same trellis, with different summaries and outputs.*

1. **Forward:** How likely is the observed journey?
2. **Viterbi:** What is the single most likely hidden path?
3. **Forward–Backward:** How likely is each state at each step?

**Takeaway:** *Compare the journey likelihood, the best hidden path, and per-step probabilities using efficient dynamic programs.*

**Why the trellis makes inference efficient**

- Different histories can reach the same state-position, e.g., $(B, t_2)$.
- Forward combines their probabilities into one summary ($\alpha_2(B)$), which is then reused in later steps.

*Arriving at $(B, t_2)$*

| From | Contribution |
|---|---|
| B at $t_1$ | $\alpha_1(B) \times A_{B,B}$ |
| E at $t_1$ | $\alpha_1(E) \times A_{E,B}$ |

*Sum, then multiply by $b_B(o_2)$*

**Without reuse**
Enumerate all hidden paths: $O(N^T)$

**With a trellis**
Compute summaries at each (state, time): $O(TN^2)$

### 1 Forward: Score the Observed Journey

*Sum the probability of all hidden paths that could produce the observations*

**Idea:** $\alpha_t(j)$ is the total probability of all partial hidden paths that end in state $j$ at time $t$ and explain the observations up to time $t$.

$$\alpha_t(s) = b_s(o_t) \sum_{s'} \alpha_{t-1}(\hat{s}) \; A_{\hat{s},s}$$

Emission (likelihood of $o_t$ in state $s$) · Previous $s$-values (total for $\hat{s}$) · Transition (from $\hat{s}$ to $s$)

**How Forward works**

1. Initialize at Search ($t_1$)
   B: $0.80 \times .50 = .400$ · E: $.20 \times .10 = .020$
2. Update to both states at View ($t_2$)
   *Reusing: add incoming mass, then emit View*
   $\alpha_2(B) = .30[(.400 \times .80) + (.020 \times .20)]$
   $= 0.0972$
   *Similarly for state E, different emission:*
   $\alpha_2(E) = .40[(.400 \times .20) + (.020 \times .80)]$
   $= 0.0384$
3. *Repeat for both states at $t_3$ and $t_4$.*

| | $t_1$ | $t_2$ | $t_3$ | $t_4$ |
|---|---|---|---|---|
| B | 0.400 | 0.0972 | 0.0171 | 0.0056 |
| E | 0.020 | 0.0384 | 0.0225 | 0.0094 |

**Worked example - Forward table**

*Observed journey: Search → View → Compare → View · States: B, E*

| State | $t_1$ Search | $t_2$ View | $t_3$ Compare | $t_4$ View |
|---|---|---|---|---|
| B | 0.4000 | 0.0972 | 0.0171 | 0.0056 |
| E | 0.0200 | 0.0384 | 0.0251 | 0.0094 |

**4. Final:** $0.00560592 + 0.00939264 = 0.01499856$ (~1.50%)

*Both final states are possible; sum unrounded values to score the journey.*

### 2 Viterbi: Find the Single Most Likely Hidden Path

*Keep one strongest complete hidden-state path.*

**Idea** Keep only the best partial path, its predecessor, then trace back.

$$\delta_t(s) = b_s(o_t) \max_{s'} \delta_{t-1}(s') A_{s',s}$$
$$\psi_t(s) = \arg\max_{s'} \delta_{t-1}(s') A_{s',s}$$

- Viterbi score: best partial-path probability to reach state $s$ at time $t$.
- Backpointer: which predecessor $s'$ gives the winning score for state $s$.

***Max vs. argmax:*** ▸ max: the best incoming contribution (how strong the best path is). ▸ argmax: the predecessor that supplied it (which path to follow).

**Why keep only the best?**
If two partial paths reach the same state at the same time, then, when identical future choices, so the weaker path can never overtake the stronger one.

**Worked computation: $t = 2$ (View)**

1. Initialize ($t = 1$):
   $\delta_1(B) = 0.4000$ · $\delta_1(E) = 0.0200$
2. Compute both states at $t = 2$ (observation = View):

For B:
$\delta_2(B) = 0.30 \max(0.400 \times 0.80, 0.020 \times 0.20)$
$= 0.30 \times \max(0.320, 0.004)$
$= \mathbf{0.0960}$ · $\psi_2(B) = B$

For E:
$\delta_2(E) = 0.40 \max(0.400 \times 0.20, 0.020 \times 0.80)$
$= 0.40 \times \max(0.080, 0.016)$
$= \mathbf{0.0320}$ · $\psi_2(E) = B$

3. Repeat for $t = 3$, 4 using the same steps.

**Worked Example: *Search, View, Compare, View***

| | $t = 1$ | | $t = 2$ | | $t = 3$ | | $t = 4$ | |
|---|---|---|---|---|---|---|---|---|
| State $s$ | $\delta_1(s)$ | $\psi_1(s)$ | $\delta_2(s)$ | $\psi_2(s)$ | $\delta_3(s)$ | $\psi_3(s)$ | $\delta_4(s)$ | $\psi_4(s)$ |
| B | 0.4000 | – | 0.0960 | B | 0.0154 | B | 0.0037 | B |
| E | 0.0200 | – | 0.0320 | B | 0.0128 | E | 0.0041 | E |

**Trace-back: reconstruct the best path**

1. Select best final state: $E$ ($t = 4$, $\delta_4(E) = 0.0040096$).
2. Follow backpointers backward:
   $B_4 \leftarrow (B_3) \leftarrow E_2 \leftarrow E_1$
3. Read the recovered path (chronological order):
   $B_1 \rightarrow E_2 \rightarrow E_3 \rightarrow E_4$

**Key Takeaways**

- Viterbi returns *one full path* (the single most likely sequence of hidden states), not uncertainty at each time step.

**Most likely path:**
$B \to E \to E \to E$

**Joint path score:**
**0.004096**

## How certain is the best complete path?

*The observations make this path more likely, but it is not certain.*

*Selected path:* $Q^* = B \to E \to E \to E$

| Path prior $P(Q^*)$ | | Observation likelihood $P(O \mid Q^*)$ | | Joint probability $P(Q^*, O)$ |
|---|---|---|---|---|
| **10.24%** | × | **4.00%** | = | **0.004096** |
| $= 0.80 \times 0.20 \times 0.80 \times 0.80$ | | $= 0.50 \times 0.40 \times 0.50 \times 0.40$ | | $= 0.1024 \times 0.0400$ |
| Probability of this hidden-state path (before seeing actions) | | Probability of the observed actions if this path is correct | | Probability of this path and the observed actions |

**Posterior** *(after seeing the actions)*

$$P(Q^* \mid O) = \frac{0.004096}{0.01499856} \approx \mathbf{27.3\%}$$

*Normalize by the Forward likelihood* $P(O \mid \lambda) = 0.01499856$.

**What it means**

- The observed actions raise the probability of this path from 10.24% to 27.3%.
- Other paths still account for 72.7% of the probability.

### 3 Forward-Backward: State Probabilities at Each Step

*Combine past and future evidence to estimate state uncertainty.*

**Why Forward-Backward?**

Viterbi finds one most likely complete path ($B \to E \to E \to E$).

*But we often want something different: How likely was the shopper in a particular state, such as Browsing, at the View in step 2 ($t_2$)?*

**Two complementary views of the same hidden moment**

***Forward $\alpha_2(B)$***
*Evidence from the past together with state $B_2$:*
$\alpha_2(B) = P(o_1, o_2, q_2 = B)$
$= \mathbf{0.0972}$

$B_2$ (Browsing) at $t_2$

***Backward $\beta_2(B)$***
*Likelihood of the future observations given state $B_2$:*
$\beta_2(B) = P(o_3, o_4 \mid q_2 = B)$
$= \mathbf{0.0892}$

*Why combine them? Once $B_2$ is fixed, the HMM makes the past ($o_1, o_2$) and future ($o_3, o_4$) conditionally independent, so we can multiply their probabilities.*

**Our running example: a shopper's journey**

| $t$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|
| $o$ (obs) | Search | View | Compare | View |
| $q$ (hidden) | B (Browsing) | E (Evaluating) | | |
| | *(e.g., exploring)* | *(e.g., comparing)* | | |

### Work backward by summarizing future possibilities

**The backward recurrence: what it does**

At time $t$ in state $s$, the remaining observations can be generated by moving to any next state $s'$:

$$\beta_t(s) = \sum_{s'} A_{s,s'} \cdot b_{s'}(o_{t+1}) \cdot \beta_{t+1}(s')$$

- $A_{s,s'}$: transition to next state $s'$
- $b_{s'}(o_{t+1})$: likelihood of next observation
- $\beta_{t+1}(s')$: likelihood of all later observations

*For each possible next state, multiply these three factors (a route) and add them, because either route could generate the remaining observations.*

**Step 1: initialize and compute $\beta_3$**

- Initialize at end: $\beta_4(B) = \beta_4(E) = 1$
- At $t_3$, only the final View remains. Sum the possible next states for each current state.

B: $B_4: 0.8000 \times 0.3000 \times 1 = 0.2400$
$E_4: 0.2000 \times 0.4000 \times 1 = 0.0800$
$\beta_3(B) = \mathbf{0.3200}$

E: $B_4: 0.2000 \times 0.3000 \times 1 = 0.0600$
$E_4: 0.8000 \times 0.4000 \times 1 = 0.3200$
$\beta_3(E) = \mathbf{0.3800}$

**Step 2: step back to $t_2$ (reuse $\beta_3$)**

Now the remaining observations are Compare and View. Use the $\beta_3$ scores (which already include the final View):

$\beta_2(B) = 0.8000 \times 0.2000 \times 0.3200$
$+ 0.2000 \times 0.5000 \times 0.3800$
$= \mathbf{0.0892}$

**The bigger picture**

- Each step backward *summarizes futures* by one more observation: $\beta_3$ covers View; $\beta_2$ covers Compare and View.
- *We reuse previously computed $\beta$ scores* (dynamic programming) *instead of enumerating every possible path.*

**Backward scores across the four observations**

| State | 1 · Search | 2 · View | 3 · Compare | 4 · View |
|---|---|---|---|---|
| B | 0.034592 | 0.089200 | 0.320000 | 1.000000 |
| E | 0.058088 | 0.164800 | 0.380000 | 1.000000 |

**Read from right to left**

At final View, $\beta_4 = 1$.
At $t_3$, $\beta$ favors E: (0.1648 > 0.0892).
Yet $\alpha \times \beta$ favors B (57.8%).

### Turn evidence into state probabilities

**Compute a posterior at $t_2$ (Browsing):**

Forward and Backward meet at the same hidden state.

*Past evidence (from Forward)*
$\alpha_2(B) = 0.0972$

AND

*Future evidence (from Backward)*
$\beta_2(B) = 0.0892$

Search, View, and be at Browsing at $t_2$.

The remaining observations after step 2 (Compare, View), assuming Browsing at step 2.

Joint support (observations AND state):
$0.0972 \times 0.0892 = 0.00867024$

Recall the probability of the entire observed journey (from the Forward pass):
$P(O \mid \lambda) = \alpha_4(E) + \alpha_4(E)$
$= 0.00560592 + 0.00939264$
$= 0.01499856$

*GIVEN* the observed journey, normalize to get the posterior:

$$\frac{0.00867024}{0.01499856} = \mathbf{57.8\%}$$

**Filtering and Smoothing: How Evidence Changes Our Belief**

| *Filtering* (prefix only) | | *Smoothing* (entire sequence) |
|---|---|---|
| Search, View | | Search, View, Compare, View |
| 0.0972 | | 0.00867024 |
| 0.0972 + 0.0384 | | 0.01499856 |
| = **71.7%** | | = **57.8%** |

Filtering normalizes the Forward scores using only Search and View. Smoothing uses the full sequence, and its 57.8% result becomes the Browsing entry at step 2 in the gamma table.

**State posteriors ($\gamma$): Smoothed probabilities at each step**

| $t$ (obs) | $P(B_t \mid O)$ | $P(E_t \mid O)$ |
|---|---|---|
| 1 (Search) | **0.923** | 0.077 |
| 2 (View) | **0.578** | 0.422 |
| 3 (Compare) | 0.365 | **0.635** |
| 4 (View) | 0.374 | **0.626** |

Gamma ($\gamma$): smoothed state posterior. Each row totals 100%; probabilities are estimates, not observed hidden states.

***Posterior decoding:***
Choose the larger probability at each step: $B \to B \to E \to E$.

### 4 Decoding: Viterbi Path vs. Posterior States

*One global path versus local state choices.*

**Viterbi**
Decoding assigns hidden-state labels to observed actions. Viterbi selects one strongest complete allowed path.

**Forward–Backward**
Forward-Backward produces gamma: smoothed state probabilities at each position. It does not select a path.

**Posterior decoding**
Posterior decoding chooses the most probable gamma state independently at every position.

*One journey, two decoded sequences*

| Observation | Search | View | Compare | View |
|---|---|---|---|---|
| Viterbi | B | E | E | E |
| Posterior | B | B | E | E |

*Only at step 2 (View)*
Viterbi: E
Posterior: B

**The individual best path**

The highest-probable complete path (Viterbi) is $B \to E \to E \to E$ with score:

**0.004096**

*It passes through Evaluating at $t_2$.*

**Aggregating over all paths**

Many complete paths pass through $B_2$ or $E_2$. Summing their probabilities gives the total support for each state at $t_2$.

$B_2$ (Browsing) · $E_2$ (Evaluating)

$t_1$ B or E · $t_4$ B or E

All paths through $B_2$: **0.00867024**

All paths through $E_2$: **0.00632832**

**From the gamma table**

State posterior: the probability of $B$ or $E$ at one step, given all four observed actions.

Posterior decoding chooses the higher chance at each step.
$B \to B \to E \to E$
At $t_2$: B 57.8%, E 42.2%.

**Comparing the evidence at $t_2$:**

| | | |
|---|---|---|
| Best complete path (Viterbi) | $B \to E \to E \to E$ | 0.004096 |
| All paths through $B_2$ (aggregate) | — | 0.00867024 |
| All paths through $E_2$ (aggregate) | — | 0.00632832 |

**The best individual path passes through $E_2$, but the total support of all paths passing through $B_2$ is greater.**

Viterbi scores one complete path. For $B_2$ or $E_2$, we instead add all paths passing through that state, dividing by $P(O)$ gives the posterior state probabilities.

**Caveat**

Posterior decoding may violate forbidden transitions. All transitions in this example are allowed.

## Chapter III — Interpreting and Applying HMMs

*Turn algorithm outputs into insight, choose the right method, and understand the limits.*

**The Analytical Workflow** — Apply the same specified HMM ($\pi$,A,B) to each observation sequence independently, then compare or aggregate the outputs.

### Three Analytical Investigations

*Use HMM inference in three complementary ways to evaluate, decode, and summarize the observed journeys.*

**1 Forward: Journey fit**

*Score how likely each observed journey is under the same HMM.*

| Journey | Observed actions ($O_1$: S=Search, V=View, C=Compare) | $P(O \mid \lambda)$ |
|---|---|---|
| $J_1$ | S → V → C → V | 0.0150 |
| $J_2$ | S → C → C → V | 0.0137 |
| $J_3$ | S → S → C → C | 0.0157 |

- Compare likelihoods for journeys of the same length under the same model.
- For different lengths, compare per-observation log-likelihood (total log $P(O) \div T$).
- Lower likelihood means the observations are less likely under this model, not automatic proof of an anomaly.

**2 Viterbi: Behavior motif**

*Decode the most likely hidden state sequence and look for behavior patterns across journeys.*

| Journey | Most likely hidden path $s^*$ | Motif (B → E → E) |
|---|---|---|
| $J_1$ | B → E → E → E | Yes |
| $J_2$ | B → E → E → E | Yes |
| $J_3$ | B → B → E → E | No |

- *Motif definition:* $B \to E \to E$ at positions 1–3 (first three hidden states of the decoded path).
- Cohort illustration (hypothetical):

| Motif B → E → E | Purchasers | Non-purchasers |
|---|---|---|
| Present | 60 / 100 (60%) | 30 / 100 (30%) |

*An association in decoded paths, not a causal claim.*

**3 Forward–Backward: Expected occupancy**

*Use state posteriors $\gamma_t(s)$ to estimate how often a state occurs across journeys.*

$J_1$: Evaluating (E) posteriors

| $t$ | $\gamma_t(E)$ |
|---|---|
| 1 | 0.0775 |
| 2 | 0.4219 |
| 3 | 0.6354 |
| 4 | 0.6262 |

Expected E positions in $J_1$:
$= 0.0775 + 0.4219 + 0.6354 + 0.6262$
$= \mathbf{1.7610}$

| Across three journeys (illustrative): | $J_1$ | $J_2$ | $J_3$ | Total |
|---|---|---|---|---|
| | 1.761 | 2.064 | 1.387 | 5.212 |

Expected proportion in E $= 5.212 \div 12 \approx$ **43.4%**

- Sum $\gamma_t(E)$ for each journey to get expected counts (repeat for $J_2$ and $J_3$).
- The denominator is 12 total recorded positions (3 journeys × 4).
- Expected occupancy is a probabilistic estimate, not a hard count.

### Field Note · What can HMM inference really establish?

*Interpret the output, not just the answer*

**Why HMMs still matter**

A simple, transparent model that makes hidden states, transitions, and emissions explicit and interpretable.

**What inference cannot prove**

Inferred states are model-based hypotheses. They help explain patterns in the observed data but do not validate the true or only explanation.

**Practical boundaries**

- Parameter learning is separate: Forward, Viterbi, and Forward–Backward assume ($\pi$,A,B) are already specified.
- Numerical underflow: long sequences can make probabilities extremely small; use scaling or log space.
- Results depend on model structure and assumptions.

### Choose the Right Algorithm

*Match your analytical question to the method and output.*

| Analytical question | Method | Output |
|---|---|---|
| How likely is the observed journey? | Forward | $P(O \mid \lambda)$ (journey likelihood) |
| What is the most likely hidden path? | Viterbi | $s^*$ (one path) |
| How likely is each state at each step? | Forward-Backward | $\gamma_t(s)$ (state posteriors) |

### Where HMMs Still Matter

*Useful when you want interpretable hidden states and sequential structure.*

- **Speech:** modeling sequences (e.g., phonemes, words).
- **Biology:** modeling state changes (e.g., protein domains).
- **Language:** part-of-speech tagging and sequence structure.
- **Customer behavior:** modeling journeys (e.g., browse → evaluate → buy).

### When to Consider Alternatives

*HMMs are a strong, interpretable baseline, but richer models may fit better when:*

- Observations are high-dimensional (e.g., text, images, sensors).
- Dependencies span longer ranges (e.g., long-term user intent).
- Behavior is highly complex or non-Markov (e.g., changing goals, multiple overlapping patterns).

### Key Takeaways

- Run inference on each journey, then compare or aggregate the outputs.
- Forward scores likelihood, Viterbi gives a candidate path, Forward–Backward quantifies state uncertainty.
- Outputs are useful evidence under the model's assumptions—not direct observations of hidden behavior.


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Sources: Q. Mei · C. Aggarwal · Jurafsky & Martin